MILITARY SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 50 OHMS, M17/152-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-17.

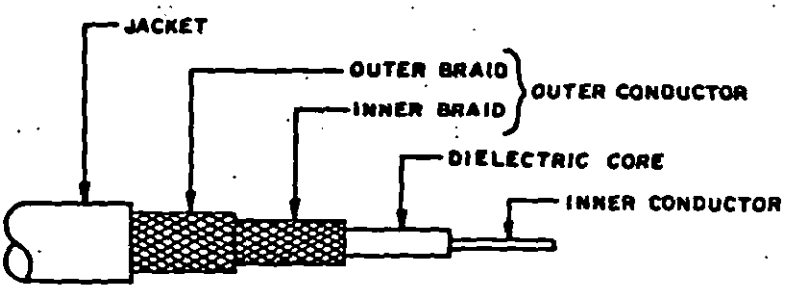


FIGURE 1. Configuration.

TABLE I. Description.

Component	Construction details
Inner conductor	Seven strands of silver-coated, annealed-copper covered steel wire, each strand .0067 inch diameter. Overall diameter: 0.0201 inch ±0.0010.
Dielectric core	Type F-1: Solid, extruded PTFE. Diameter: 0.060 inch ±.003.
Outer conductor	Double braid of AWG size 38, silver-coated, copper wire. Diameter: .096 inch maximum.
Inner braid	Coverage: 95.4%, nominal. Carriers: 16 Ends: 5 Picks/inch: 12.0 ±10%
Outer braid	Coverage: 94.6% nominal Carriers: 16 Ends: 6 Picks/inch: 12.1 ±10%
Jacket	Type IX. Diameter: .114 inch ±.004.

(A) denotes changes

FSC 6145

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Engineering information:

Continuous working voltage: 900 V rms, maximum.

Operating frequency: 12.4 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -55°C to +200°C.

Inner conductor properties:

DC resistance (maximum at 20°C): 8.41 ohms per 100 feet.

Elongation: 10 percent, minimum.

Tensile strength: 50 $klb_f/inch^2$, minimum.

Engineering note: This cable useful in general purpose, high temperature applications (see connector series "SMA", "SMB", and "SMC" per MIL-C-39012.)

REQUIREMENTS:

Dimensions, configuration, and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical:

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 2 pounds minimum, 8 pounds maximum.

Aging stability: Not applicable.

Stress crack resistance: +230° ±5°C.

Outer conductor integrity: Not applicable.

Cold bend: -55°C ±2°C.

(A) Dimensional stability: +200°C ±5°C.

Inner conductor from core: 0.125 inch, maximum.

Inner conductor from jacket: 0.197 inch, maximum.

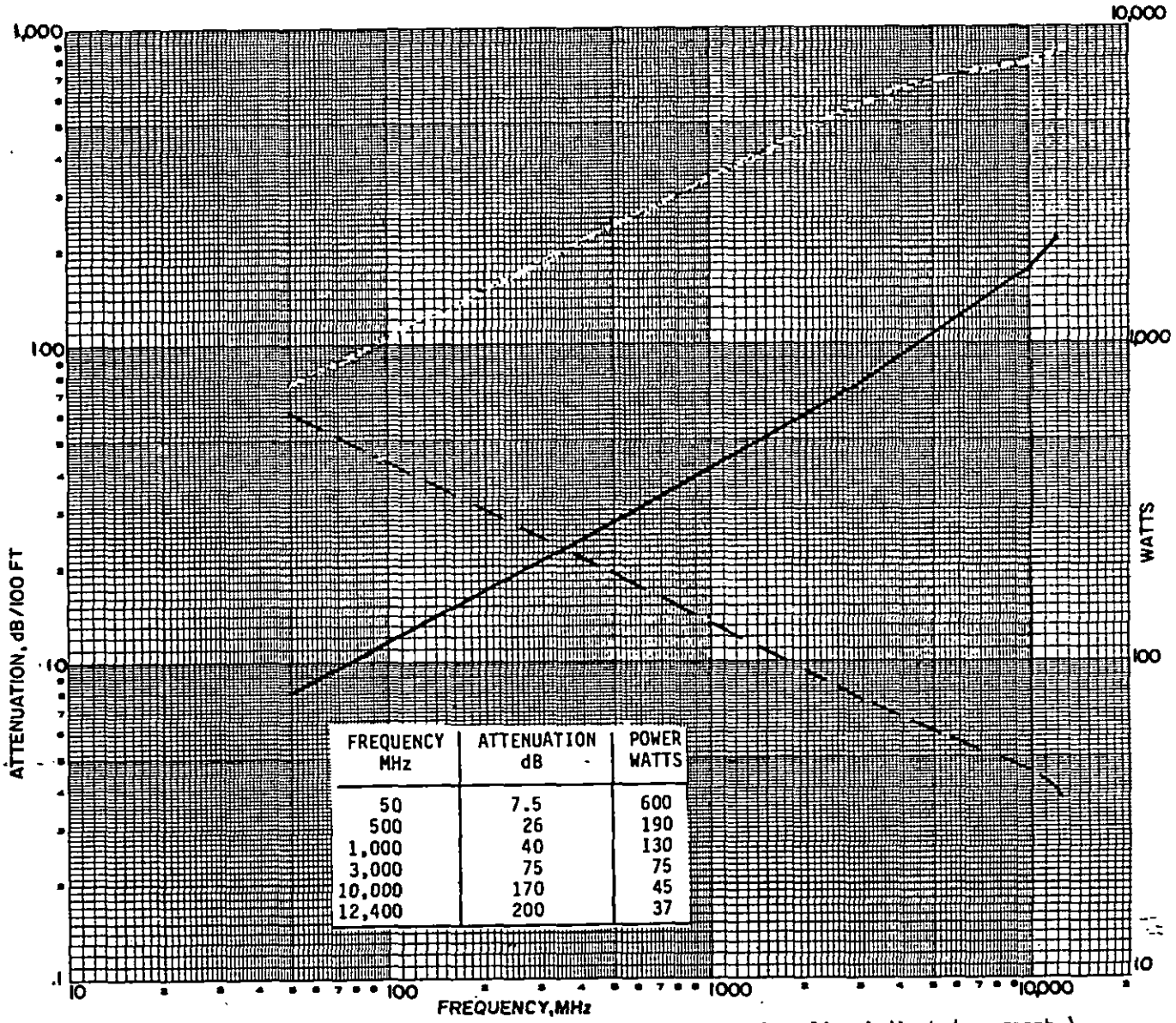
Contamination: Not applicable.

Bendability: Not applicable.

Flammability: Not applicable.

Weight: 1.85 pounds per 100 feet, maximum.

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FREQUENCY MHz	ATTENUATION dB	POWER WATTS
50	7.5	600
500	26	190
1,000	40	130
3,000	75	75
10,000	170	45
12,400	200	37

MAXIMUM ATTENUATION —— (Test requirements shall be as noted as line indicated on graph.)
MAXIMUM POWER – – –
AT 25°C SEA LEVEL

Tabulated values are for reference only.
The values on the chart represent the requirements. .

FIGURE 2. Power rating and attenuation.

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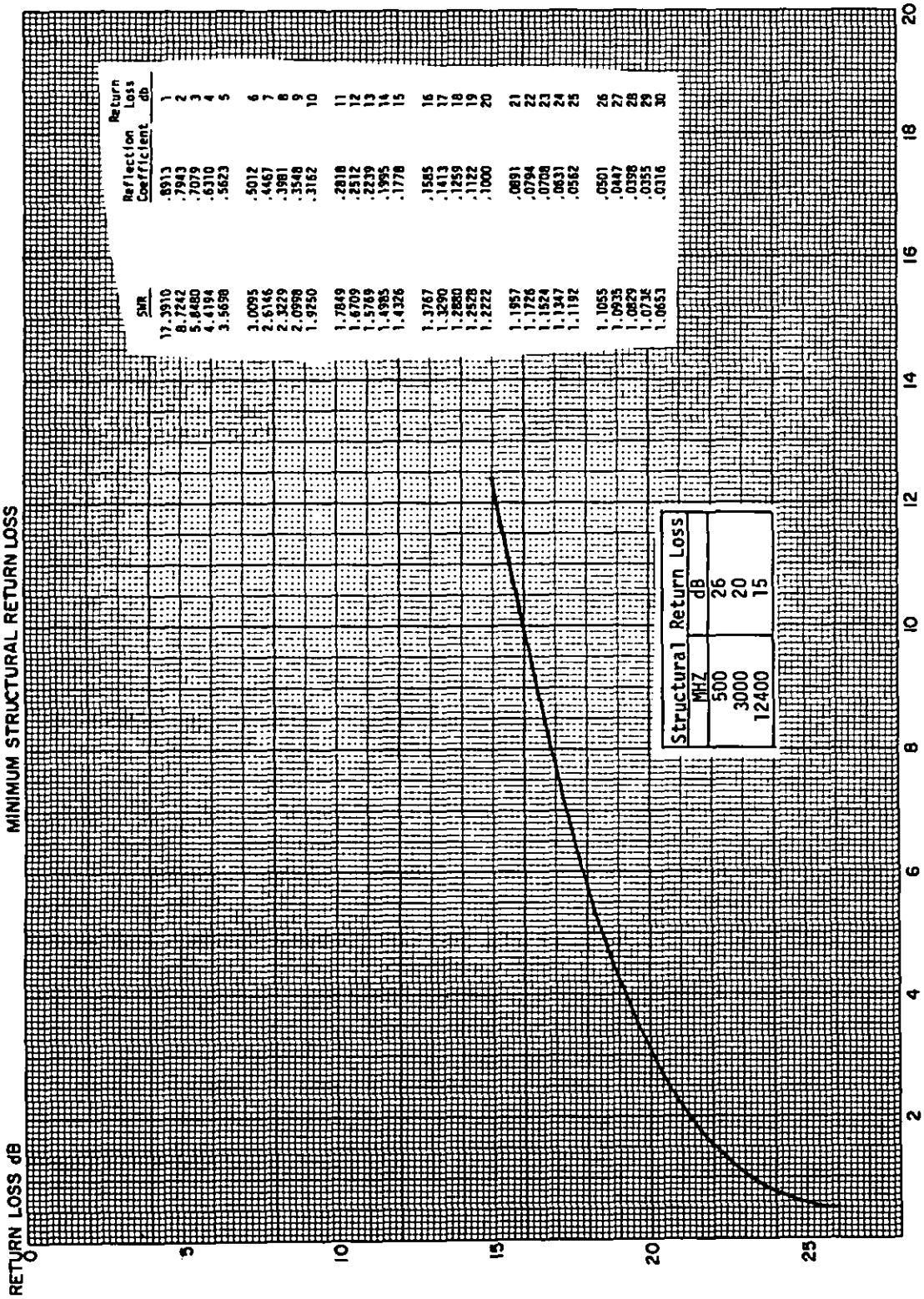


FIGURE 3. Structrual return loss.

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Electrical:

Continuity: Applicable.

(A) Spark test: 2,000 V rms +10%, -0%.

(A) Voltage withstanding: 2,000 V rms +10%, -0%.

Insulation resistance: Not applicable.

Corona extinction voltage: 1,200 V rms, minimum.

Characteristic impedance: 50 ohms ±2.

Attenuation: See figure 2.

Structural return loss: See figure 3.

Capacitance: 32 pF per foot, maximum.

Capacitance stability: Not applicable.

Capacitance unbalance: Not applicable.

Transmission unbalance: Not applicable.

Mechanically induced noise voltage: Not applicable.

Time delay: Not applicable.

Part number: M17/152-00001.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, TD
Air Force - 11, 17, 99
DLA - ES, IS

User activities:
Army - AR, AT, ME
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - CR

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